TeleOracle: Fine-Tuned Retrieval-Augmented Generation With Long-Context Support for Networks

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***Abstract*—The telecommunications industry's rapid evolution demands intelligent systems capable of managing complex networks and adapting to emerging technologies. While large language models (LLMs) show promise in addressing these challenges, their deployment in telecom environments faces significant constraints due to edge device limitations and inconsistent documentation. To bridge this gap, we present TeleOracle, a telecom-specialized retrieval-augmented generation (RAG) system built on the Phi-2 small language model (SLM). To improve context retrieval, TeleOracle employs a two-stage retriever that incorporates semantic chunking and hybrid keyword and semantic search. Additionally, we expand the context window during inference to enhance the model's performance on open-ended queries. We also employ low-rank adaption for efficient fine-tuning. A thorough analysis of the model's performance indicates that our RAG framework is effective in aligning Phi-2 to the telecom domain in a downstream question and answer (QnA) task, achieving a 30% improvement in accuracy over the base Phi-2 model, reaching an overall accuracy of 81.20%. Notably, we show that our model not only performs on par with the much larger LLMs but also achieves a higher faithfulness score, indicating higher adherence to the retrieved context.**

***Index Terms*—6G networks, AGI, large language model (LLM), low-rank Adaptation (LoRA), retrieval-augmented generation (RAG).**

I. Introduction

THE RECENT interest in large language models (LLMs), motivated by their unprecedented performance on various downstream tasks, has led to their widespread adoption across many domains. These models gained notable attention with a rising number of applications in text summarization, classification, and generation. In telecom, the potential gain from the incorporation of these models has not gone unnoticed [2], [3], [4], [5], [6], [7]. The integration of LLMs into IoT devices represents a crucial step toward creating autonomous, agentic systems capable of making real-time decisions at the network edge [8]. However, the integration of language model (LM) to domain-specific edge use-cases poses several challenges that is yet to be addressed.

Foundation LLM models, trained on vast amounts of diverse data, offer versatility and exhibit strong performance across generalized tasks. However, this broad knowledge can result in so-called negative transfer or knowledge interference in specialized domains, such as telecommunications, where domain-specific terms and concepts often conflict with their general usage. Indeed, protocols and concepts in telecom sometimes follow distinct logic patterns that differ from broader contexts. This misalignment leads to poor performance when foundation LLM models, such as GPT4, attempt to reconcile their general knowledge with telecom-specific concepts, as the former can actively interfere with the latter. In addition, the strict latency requirements and resource constraints of edge devices, particularly in massive IoT deployments, hinder the deployment of LLMs, necessitating a shift toward smaller models [9]. Additionally, there are several bodies that define telecom standards and frameworks, including 3GPP and O-RAN. Telecom documents in both ORAN and 3GPP are highly specialized and often contain terminology and concepts unique to each framework. Semantic routers [10] can improve the contextual relevance by routing queries based on their content to the suitable database (DB).

Small language models (SLMs), such as Microsoft's `Phi-2` with 2.7-B parameters [11] and `Gemini Nano 2` with 3.2-B parameters [12], present a potential solution by offering competitive performance while maintaining the efficiency crucial for telecom applications. Notably, despite its smaller size, Phi-2 performs on par with state-of-the-art LMs 25 times its size on various benchmarks. Telecom is a rapidly evolving domain that is continually adapting to accommodate the fast-paced technology development. As a result, technical papers and documents describing new ideas, standards, and protocols undergo frequent modifications, requiring artificial intelligence (AI) systems developed for the domain to be equally flexible

Received 2 February 2025; revised 6 March 2025; accepted 13 March 2025. Date of publication 20 March 2025; date of current version 9 May 2025. This article was presented in part at the 2024 IEEE Global Communications Conference (Globecom) [DOI: 10.48550/arXiv.2408.11775]. *(Corresponding author: Omar Alhussein.)*

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Digital Object Identifier 10.1109/JIOT.2025.3553161

in updating their knowledge. This limits the utility of using fine-tuning to specialize the model to the domain. Unlearning in LLMs is also a significant challenge [13]. Therefore, a more dynamic and adaptable approach, such as retrieval-augmented generation (RAG) is needed for greater flexibility and cost-effectiveness. This approach enables the LMs to seamlessly incorporate new data without the need for extensive retraining.

RAG minimizes hallucinations by grounding model responses in factual information through a two-step process: 1) storing domain-specific documents in a DB (in semantic space), and 2) supplementing the user's query with relevant retrieved context at inference time. The retrieved context can also include user-specific information or real-time system updates, allowing the model to leverage the user interaction history to produce tailored responses and remain aligned with current data. For these reasons, RAG is particularly well suited for telecom applications.

In this article, we aim to develop an advanced specialized RAG model to address the unique challenges in the telecommunications domain. Telecom documents come with varying structures, which present the first challenge. Moreover, many recurring terms have context-dependent interpretations. This means retrievers must meticulously sift through and select the most pertinent document chunks to provide the generative model with precisely targeted contextual information. Accurately retrieved information serves as the foundation for generating an accurate and effective response. Moreover, open-ended or vague user queries require a larger number of retrieved context chunks. However, SLMs have a limited context window, reducing their ability to handle larger input sequences required for more complex and open-ended questions. To address the aforementioned challenges, this article makes the following contributions to specialized LMs for telecommunications.

1) We develop TeleOracle, a specialized RAG framework that effectively addresses the unique challenges of processing telecommunications documentation through strategic integration of multiple techniques.
2) We implement an optimized document processing pipeline that combines semantic chunking with a two-stage retrieval process, enabling precise and context-aware selection of relevant technical information.
3) We leverage SelfExtend to extend the generator's context window at inference time, accommodating a greater volume of retrieved information and enabling more comprehensive responses to complex or open-ended queries.
4) We leverage a semantic router to route queries to the suitable telecom standards and frameworks DB. By utalizing the semantic router we can ensure that the query is processed with the most relevant context, improving the accuracy of the response.
5) We conduct a comprehensive analysis of the proposed architecture. Notably, we show that our model not only performs on par with state-of-the-art LLMs models but also is higher on the faithfulness score, indicating higher adherence to retrieved context.

The remainder of this article is organized as follows [13]. Section II presents the relevant works. Section III provides a comprehensive overview of the TeleOracle architecture. Our experiments and their corresponding results are detailed in Section IV. Finally, Section V concludes this article with a summary of findings and potential directions for future research.

II. Related Works

With growing interest in integrating LLMs into telecom, the need for datasets for benchmarking has become increasingly crucial. To this effect, Karim et al. [14] introduce a dataset, *SPEC5G*, composed of scraped Third Generation Partnership Project (3GPP) documents tailored for standards analysis tasks. From the created *SPEC5G* dataset, they curate two annotated subsets to evaluate the effectiveness of *SPEC5G* for downstream tasks. To test the effectiveness of their dataset, they pretrain a model on the *SPEC5G* dataset and fine-tune it on one of the curated datasets for downstream tasks. They find that the performance of the pretrained model surpasses that of the baseline models.

Nikbakht et al. [15] presents the *TSpec-LLM* dataset. This dataset covers 3GPP documents from releases 8 to 19. The authors also develop an automated framework to generate question and answer (QnA) pairs. They use a RAG model to assess the quality of the dataset and find that supplementing the LLM with context from the dataset significantly enhances the model's performance. Focused on open radio network (O-RAN) specifications, Gajjar and Shah [16] release *O-RAN-Bench-13K*. This benchmark dataset is generated through a three-stage LLM question multiple-generation system from the O-RAN specification documents. They also develop ORANSight, a RAG framework, and find that this framework enhances the performance of LLM on the O-RAN QnA dataset.

Several works focus on the creation of a QnA telecom benchmarking dataset. Holm introduces *TeleQuAD*, a private benchmark dataset generated for telecom QnA tasks [17]. The dataset consists of 4 000 questions and answers generated from randomly sampled sections of telecom documents. Similarly, the *TeleQnA* dataset emerges as a comprehensive contribution to telecom QnA work, encompassing 10 000 entries carefully curated from various sources, including 3GPP specifications and research papers [18]. They also present a framework for automated generating of QnA entries that relies on LLMs and human validation. They evaluate the performance of GPT-3.5 and GPT-4 on the *TeleQnA* dataset with and without added context in the prompt. Their results confirm the need for a specialized telecom LM and the utility of supplementing the prompt with context to better align the models. Furthermore, they evaluate their models against professionals in the field and find that the LM perform competitively with humans.

Driven by the LLM's potential, several works explore the adaptation of specialized LLMs for telecom tasks. Ahmed et al. [19] compare retrained models to fine-tuned models to evaluate the performance of LLM in telecommunications tasks. On the other hand, Bariah et al. [20] focus on fine-tuning LLMs for text classification purposes. Another notable example is TeleRoBERTa, developed by Ericsson. TeleRoBERTa is a question-answering LLM model, that is fine-tuned on a substantial corpus of telecom-specific texts, particularly 3GPP specifications [21]. These works highlight the potential for these models in the telecom domain.

RAG models have also been an area of interest in the field. *Telco-RAG* is a RAG model optimized for telecom applications [22]. Their work focuses on conserving random access memory and enhancing the prompt and user's query for better retrieval. They leverage a router to determine the most fitting 3GPP series to retrieve from and use candidate answer generation to enhance the retrieved chunks. Similarly, Yilma et al. [23] use RAG to align the LLM to the telecom domain. Yilma et al., modify the prompt by complementing the user's query with information from their previous queries. Furthermore, they use role-playing prompting techniques to address the user's query in the appropriate depth and complexity.

SLMs have emerged as a notable focus in telecom. Piovesan et al. [24] emphasize the necessity of this shift toward smaller models, to overcome edge computing challenges. They perform an analysis of the performance of Phi-2 on a downstream telecom QnA task and find that RAG notably enhances the model's performance underscoring its potential in such tasks. Guda et al. [25] introduce techniques, such as as diversified embedding models, and option shuffling to enhance the performance of small language models in answering telecom multiple-choice questions. Similarly, Lee et al. [26] analyzes the domain-specific requirements and proposes a SLM-based RAG framework They employ advanced prompting and retrieval techniques to enhance the performance of their RAG model for telecom multiple choice question (MCQ). Recently, Maatouk et al. [27] present Tele-LLMs, a collection of telecom-specialized LLMs. They also introduce two telecom datasets, Tele-Data and Tele-Eval, for training and evaluation purposes. Roychowdhury et al. [29] investigate the effectiveness of some of the RAG assessment [28] metrics on a telecom QA dataset. They determine that faithfulness and answer-correctness, which evaluate the relevance of the context and the accuracy of the output, respectively, are highly vital in telecom.

It is worth mentioning that efforts to automate network operations has predated the emergence of LLMs, such as with trigger-action programming (TAP)-based systems. The growing complexity of network operations, combined with the need for flexible policies and increasing user demands, drives this shift toward automation. TAP-based systems allows users to define simple automation `if-then` automation rules, making them particularly valuable in IoT environments where numerous devices need to interact seamlessly. However, not all users have technical knowledge and therefore do not fully benefit from the conveniences brought on by TAP. Some works have proposed a rule recommendation system that addresses this issue using techniques, such as semantic-aware approaches, heterogeneous-graphs, and graph-contrastive-learning [30], [31], [32]. Although the TAP paradigms enhance the user's interaction with the network and enable the automation of various aspects of network management, they fail to fully align with the evolving demands of the network. LLMs can be more suitable for these environments that involve unstructured data, such as conditions and actions that are not clearly defined. Furthermore, LLMs potential to leverage multimodal data enables the system to develop a more comprehensive understanding of the network, resulting in better decision making by incorporating diverse sources of information and adapting to dynamic conditions. The work presented by Cheng et al. [33] implements a hybrid LLM-TAP approach to automate IoT operations while making them more accessible to nontechnical users, effectively bridging the gap between sophisticated network automation and user-friendly interfaces.

While previous works demonstrate the potential of RAG [22], [23] and highlight the importance of SLMs for edge computing [24], TeleOracle introduces new contributions that address key challenges in telecom documentation processing. Unlike existing approaches, we implement semantic chunking and a two-stage retrieval process with a hybrid retriever and reranker to handle the contextual complexity of telecom terminology and concepts. Furthermore, by combining SLMs with context window extension, we enable comprehensive responses to complex queries while maintaining the efficiency benefits of smaller models. We also incorporate a semantic router that directs queries to the appropriate DB, ensuring that the most relevant context is retrieved for each query. Our evaluation shows that TeleOracle not only matches larger models' performance but achieves higher faithfulness scores, making it particularly suitable for technical telecom applications.

III. Proposed Architecture

A. General Overview

TeleOracle's RAG architecture is composed of four integral components: 1) a semantic document splitter to chunk the documents; 2) an embedding model to encode text to vector representations; 3) a two-stage retriever for context selection; and 4) an SLM generator (Phi-2) to produce the responses. The entire TeleOracle architecture is illustrated in Fig. 1.

In the proposed RAG framework, the generator, Phi-2, is efficiently fine-tuned using LoRA. Furthermore, TeleOracle's setup involves populating a DB with semantic chunks of the context documents (detailed in Sections III-B and III-C). After a query is passed to TeleOracle, the appropriate DB to be used by the retriever is determined by the router. Then the query is forwarded to the two-stage retriever, consisting of the hybrid retriever and reranker (described in Section III-D). This context is then incorporated into a prompt alongside the user's query, specific instructions for Phi-2, and answer options. To expand the context window and enable the LM to handle input sequence beyond the length encountered during fine-tuning, we use SelfExtend, outlined in Section III-E. In what follows, we elaborate on each of these components.

B. Semantic Chunking Strategy

The varying structures of the telecom documents compromise the quality of the DB when naive chunking mechanisms

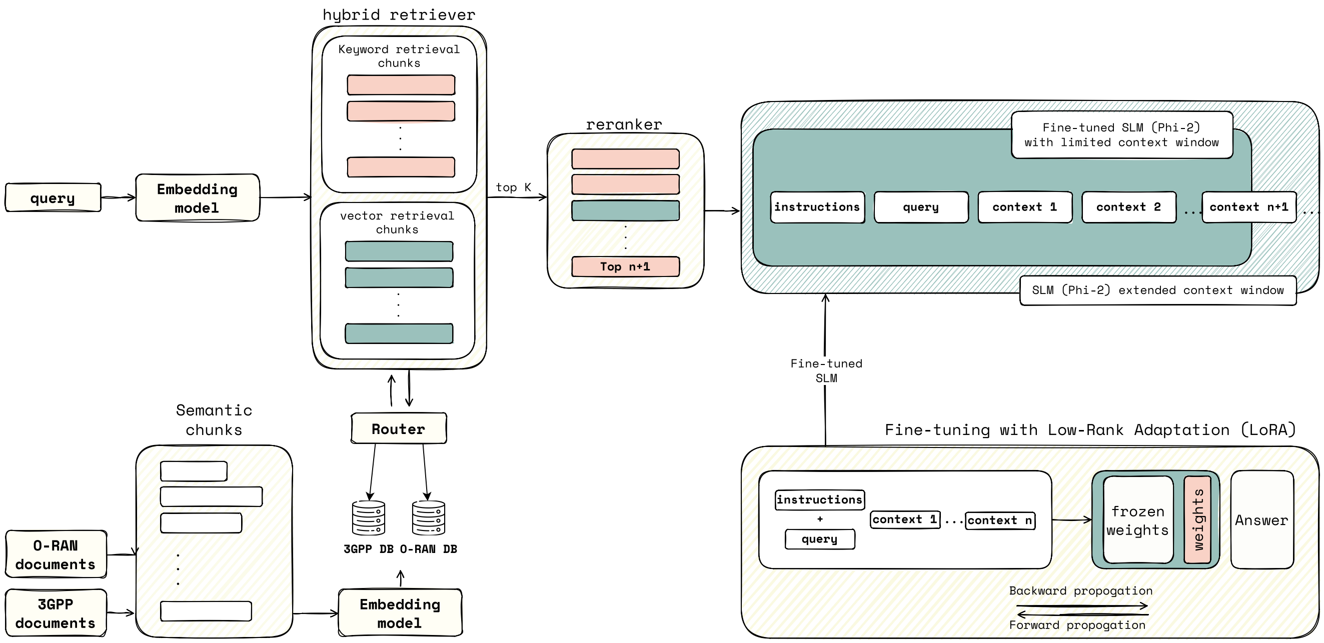


Fig. 1. Overview of TeleOracle architecture with semantic chunking, two-stage retrieval process, extended context support at inference time, and fine-tuned Phi-2 SLM integration with LoRA.

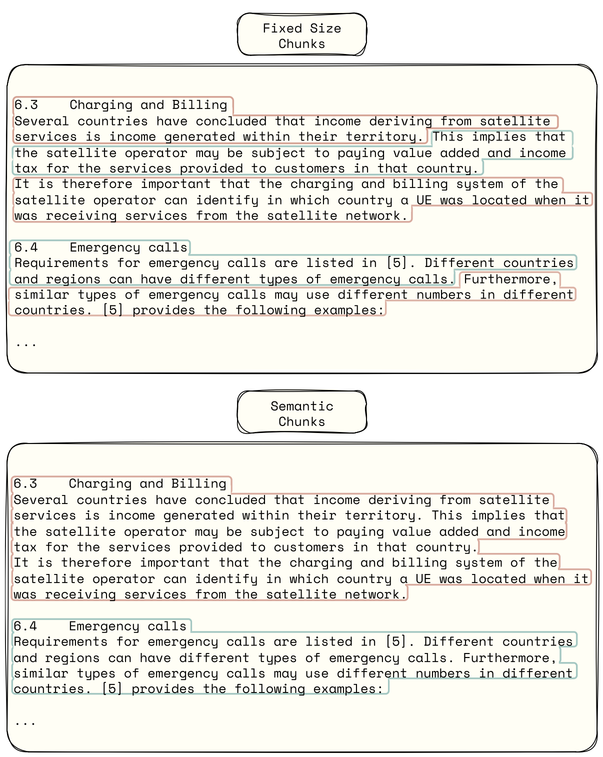


Fig. 2. Comparison of fixed-size chunking and semantic chunking applied to an excerpt from a 3GPP document.

are used. This is because the effectiveness of RAG's retrieval mechanism heavily depends on chunking - the process of breaking documents into smaller segments - as it determines not only how information is organized within the embedding system but also how precisely and efficiently relevant context can be retrieved during queries.

One of the common chunking methods is fixed-size chunking, which involves splitting the chunk based on a fixed chunk length. This method allows the size and number of chunks to be predetermined to fit the type of documents and the generator's input size limitations. However, the documents need to be of similar format and a careful study needs to be conducted to determine the best chunk size. In contrast, semantic chunking offers greater flexibility by dynamically determining split points based on a dissimilarity threshold. In this work, we utilize semantic chunking to enhance the relevance and accuracy of the retrieved information. This technique involves splitting the text whenever a set dissimilarity threshold is exceeded. By grouping sentences that are close in the embedding space, we ensure that each chunk is coherent, capturing contextually related information together. Fig. 2 compares these two approaches, illustrating how semantic chunking better maintains contextual integrity across document segments.

C. Embedding Model and Chroma DB

To enable LM to process a sequence of text effectively, each element in the sequence must be assigned a numerical representation that captures the semantic meaning and the relationships between elements. These are known as the vector representations of these elements and are created using the embedding model. They are used to evaluate the similarity between the elements. In this article, we use *bge-small-en-v1.5* [34], which is optimized for limited resource environments that benefit from a small efficient model. The model is trained using contrastive learning [35], a technique that entails minimizing the distance between semantically

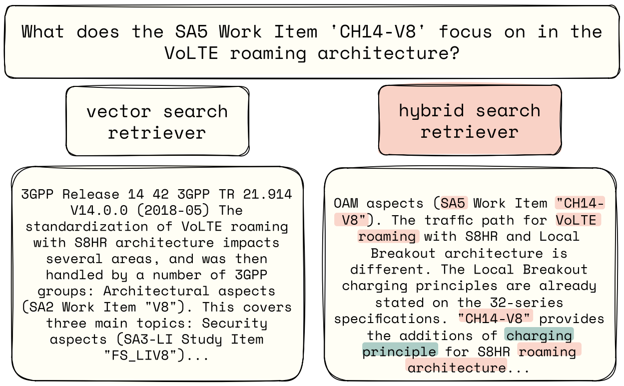


Fig. 3. Comparison between vector search and hybrid search retrievers. The retrieved text for the question "What does the SA5 Work Item "CH14-V8" focus on in the VoLTE roaming architecture?" is shown. The answer to the question is highlighted in blue.

similar pairs while simultaneously maximizing the distance between dissimilar pairs in the embedding space.

Computing these embeddings is time-consuming, thereby making the use of a vector DB essential for faster performance at runtime. Traditional DBs lack optimization for vector-based similarity searches, which involve finding nearest neighbors (i.e., the most similar vectors). This process can be computationally intensive and time-consuming, particularly when handling large datasets. Vector DBs, designed specifically for managing high-dimensional embeddings, offer significant performance advantages in these scenarios. This makes Chroma DB an advantageous choice. This DB is an open-source AI-native vector database specifically designed for managing high-dimensional embeddings. Its dual support for local and remote storage enables organizations to either maintain strict data privacy through local control or leverage cloud-based scalability for edge device deployments. This flexibility enables customized implementations, striking an optimal balance between robust security controls, system performance, and horizontal scalability to precisely match their application's unique requirements.

Furthermore, a semantic router is used to route user queries to the appropriate DBs. The router works by embedding various examples of queries from each category, then semantic similarity is used to determine the best routing decision. We optimize the similarity threshold based on a dataset with MCQs based on 3GPP and O-RAN documents by iteratively adjusting the threshold. As opposed to LLMs, semantic routers provide faster and more deterministic decisions. Additionally, by reducing unnecessary LLM calls, semantic routing can lower operational costs. These routers can also be used to provide guardrails, preventing irrelevant or risky queries from being processed by the LLM.

D. Two Stage Retrieval

RAG's performance hinges on its ability to ground generator responses in accurate and relevant context. Our work addresses this through a two-stage retrieval process: a hybrid search retriever paired with a reranker.

1) Hybrid Search Retriever: We implement a hybrid retriever that leverages the strength of both semantic and keyword-based search to enhance context retrieval.

Keyword search retrievers are used to find chunks based on specific words or phrases. Having a keyword-based retriever is essential for telecom applications because of the presence of telecom-specific terms, such as protocol or document names that must be accurately identified and retrieved. For instance, in Fig. 3, the question references several key terms, such as 'SA5' (Service and System Aspects 5), 'CH14-V8' (an identifier for a specific work item within the SA5), VoLTE (Voice over LTE, and roaming architecture. These terms are crucial for understanding the context and accurately addressing the query.

In this work, we use BM25 [36] as our keyword search retriever, which scores queries as follows:

$$\text{BM25\_score} = \frac{f(q, D) \times (k+1)}{f(q, D) + k \times \left(1 - b + b \times \frac{dl}{adl}\right)} \times \log\left(\frac{N - N(q) + 0.5}{N(q) + 0.5} + 1\right) \tag{1}$$

where $f(q, D)$ is the frequency of the query term q in document D, dl is the length of document D, adl is the average document length across the collection, N is the total number of documents in the collection, and $N(q)$ is the number of documents containing the query term q. The inverse document frequency is represented as $\log(N/N(q))$. This term quantifies the rarity of the term q. The more documents the term appears in, the lower this fraction becomes, and hence the smaller the score. The parameters k and b are used to fine-tune the formula, where k controls term frequency saturation and b adjusts for document length normalization.

BM25 is needed in our case to retrieve telecom-specific terminology because it effectively ranks documents based on the presence and frequency of these specific terms. By prioritizing documents that contain these critical keywords, it ensures that the most relevant information is retrieved. This retriever improves on an existing similarity search method, namely term frequency inverse document frequency (TF-IDF). TF-IDF calculates the relevance of a term based on its term frequency and the inverse document frequency, which is the measure of how rare or common a term is across the entire corpus. This allows it to score terms that occur often in all documents, such as "The," lower compared to terms that occur often only in a few documents, such as "NWDAF."

TF-IDF identifies important terms while differentiating between common and rare terms. However, there are more factors to consider when deciding on the relevance of the document. BM25 builds on TF-IDF and addresses those factors. For instance, the frequency of a term *TF* is not directly proportional to its relevance. In other words, if the term "NWDAF" occurs 40 times in one document and 20 times in another, the document with 40 occurrences is not necessarily twice as relevant as the one with 20 occurrences. Therefore, the term k is added to moderate the influence of the term frequency on the score. Furthermore, BM25 considers the influence of the length of the document on the significance of term frequency

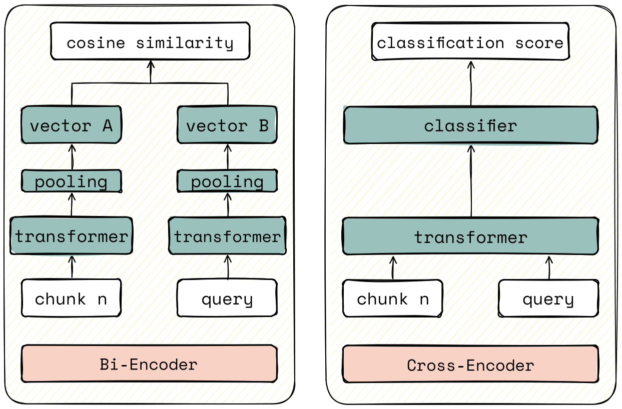


Fig. 4. Bi-Encoder versus Cross-Encoder architectures. In the Bi-Encoder, query and chunk are encoded separately, and cosine similarity is used to compare embeddings. The Cross-Encoder jointly encodes the query and chunk, producing a classification score.

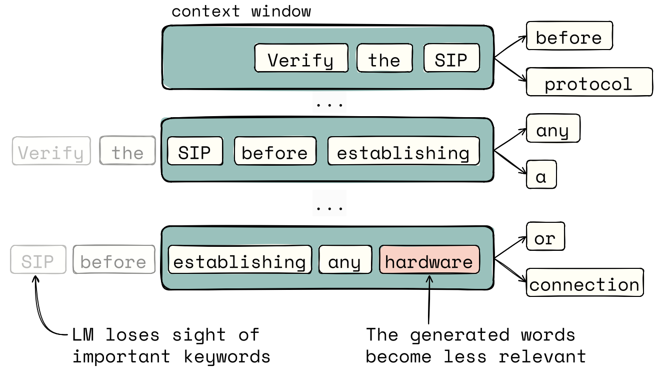


Fig. 5. Example of how language models lose important context when input sequences exceed their fixed context window. As the sequence length grows, the model forgets key tokens like "SIP" and generates less relevant outputs, such as "hardware," due to its inability to retain words beyond the context window.

TF. A term appearing twice in a short document can be more significant than a term that appears twice in a much longer document. BM25 introduces a new term that penalizes longer documents (dl/adl). These additional components render the BM25 a better fit for this work than the TF-IDF.

Vector search relies on identifying the semantic similarity between vectors, measured by the relative distances between them. The vector search retriever first vectorizes the query using the embedding model, converting it into a high-dimensional embedding. This vector captures the query's meaning. The retriever then calculates the cosine similarity between this query's vector and the existing vectors stored in the database. Cosine similarity, expressed as the dot product of two vectors over their magnitude, presents as a low-cost operation that captures the similarity between vectors. Vector search excels at understanding the context and nuances of complex queries, which is crucial for retrieving context that adequately addresses telecom-related questions.

After a list of context chunks is retrieved by the semantic and keyword-based search, a merged list is created by taking the union of both lists of retrieved chunks [37]. Hybrid search retrievers capture intricate details and words while also understanding the overall context. Fig. 3 demonstrates an example from our experiments where the hybrid retriever provides a more relevant output compared to only using a vector retriever.

2) Reranking: The first retrieval stage, the hybrid retriever, is computationally inexpensive but does not capture the fine-grained contextual relationships between the query and documents. Relevant context concealed within the middle of the contexts list tends to be overlooked by the LLM [38]. This highlights the need to refine the list of contexts. To achieve this we incorporate a cross-encoder reranker into our framework [39]. Cross encoders, illustrated in Fig. 4, evaluate the similarity between each query and document pair and then output a classification score between between 0 and 1, where 1 signifies relevance and 0 indicates irrelevance. This allows cross-encoders to capture nuances between context chunks the query, however, the pairwise calculations make this a time-consuming and computationally expensive process.

On the other hand, bi-encoders, used in stage one, encode all the documents in advance before performing a search, enabling faster and more efficient retrieval. However, they typically underperform compared to cross-encoders in terms of accuracy, as they may fail to capture complex interactions between the query and context chucks. This further highlights the tradeoff between efficient performance and accuracy and demonstrates that a two-stage retrieval, such as the one employed in this framework, is essential to maintain an acceptable inference time, while providing optimal accuracy.

3) Time-Complexity: The TeleOracle framework prioritizes efficiency without compromising accuracy, employing advanced techniques, such as hybrid retrieval and reranking to improve query-context matching. However, the inclusion of certain components introduces computational overhead, particularly in the retrieval and ranking phases. The cross-encoder, used in the reranking stage, is the bottleneck of the framework in terms of latency. Given N retrieved chunks and a query, this results in a complexity of $O(N \times M)$, where M represents the input query length of each context. The dynamic nature ensures nuanced scoring but comes at the cost of increased latency. We balance this by leveraging bi-encoders to shortlist the number of chunks for which to compute the similarity score. With precomputed embeddings, the bi-encoder can narrow down a large pool of chunks and obtain the top k chunks to feed into the cross-encoder. This two-stage approach is computationally feasible even with large DBs.

E. Extending the Context Window With SelfExtend

During training, LMs are exposed to a fixed context window length, i.e., an input sequence of a fixed length. If the context window's length during inference exceeds what the LM was trained on, it behaves unpredictably and produces gibberish outputs. This occurs because LMs cannot retain words that extend beyond the context window, this is illustrated in Fig. 5.

Though this inability to generalize may not present as an issue for larger propriety models with large context windows,

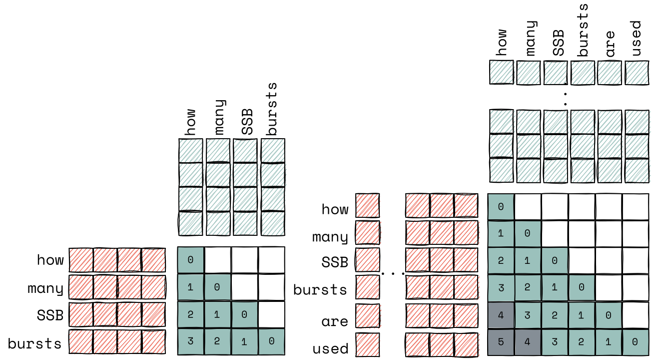


Fig. 6. Illustration of the positional Out-of-Distribution (OOD) issue. The figure demonstrates the relative distances when the input sequence length is within the pretrained model's context window (left), and how unseen relative distances arise when the input sequence exceeds the model's context window (right).

it becomes particularly problematic for smaller open-source language models. These models are trained on input small sequences due to limitations in computational resources and the available training data. Addressing this limitation is critical as chunking techniques, such as semantic chunking, and future improvements, such as adding tables to the context, the context window of the LMs will need to be expanded. Furthermore, a bigger number of chunks might be retrieved to adequately answer broad questions. For instance, "From a UE perspective, what does the UE need to be aware of?" and "How are messages routed in the 5G system?" are inherently broad questions. They could touch upon numerous aspects, such as network registration, protocol compliance, and so on. To adequately respond the generator needs to be supplemented with a substantial number of chunks.

Furthermore, in our tests, we exclude the MCQ options from the query during retrieval to simulate a more realistic user interaction when posing a question. This approach ensures that the LM does not rely on preprovided answer choices, thereby preserving the complexity of the task. Certain questions in the dataset, such as "From a BS perspective, which statement is true?" and "Which of the following is NOT a function of XnAP?", benefit from the inclusion of options as they help narrow down the retrieved text. Without the provided options, a more extensive list of retrieved-context would be necessary to answer these types of questions accurately.

It is also wrong to assume that all users of the telecom oracle are proficient in prompting LM. For instance, inexperienced users might use queries that might be vague, ambiguous, or lack the specificity needed to extract the most relevant information. Providing the LM with a wide array of context could allow it to sufficiently address the user's query.

The issue of input sequences longer than the pretrained model's context windows has been addressed in several works. Most proposed techniques involve fine-tuning in a longer context [40], [41], [42]. However, fine-tuning requires extensive resources. Jin et al. [43] propose SelfExtend, a method to extend LMs context window without requiring fine-tuning. SelfExtend alters the way in which input is passed to the self-attention mechanism by modifying the positional encoding. Self-attention is a foundational concept in transformers that involves processing tokens in the input sequence in parallel. It requires positional encoding to be applied to each token to preserve information about the word order.

In other words, the tokens are assigned values that represent their position, providing useful information to the LM about the relationship between tokens in the sequence. In a sequence of tokens, each token is assigned an absolute position. For instance, in the sequence $\{w_1, w_2, \ldots, w_L\}$ each token w_i has absolute position i. Let m represent the absolute position of the query token, and n represent the absolute position of the key token in the sequence. The difference $m - n$ is the relative position which is critical for the self-attention mechanism to understand how tokens are positioned relative to each other.

Jin et al. show that altering this encoding can unlock the LM's ability to handle longer input sequences. SelfExtend can be applied to models that use rotary position Embeddings (RoPE), one of the many to encode positional information for each token [44]. Other positional encoding methods, such as absolute positional encoding either lack the ability to generalize to unseen positions, or struggle to handle long sequences. RoPE uses a rotation matrix where the token's absolute positional encoding is rotated depending on its position in a sentence.

Though RoPE is theoretically capable of handling longer sequences, it faces challenges in practice, namely, its inability to distinguish between distant values as the rotational angles become too large, leading to a loss of positional accuracy. When the model is exposed to sequences longer than it encountered during training, these tokens are assigned positions that fall outside the distribution of the training data as illustrated in Fig. 6. This suggests that the models struggle with longer input sequences not because of an inherent lack of ability to handle long input sequences, but rather due to the model overfitting on the positional encoding it was exposed to during fine-tuning.

SelfExtend builds on the idea that relative positions do not have to be unique, and uses the floor operation resulting in position indices up to the maximum position index seen during training. This approach, where a few tokens are mapped to the same position embedding, is referred to as grouped attention. The relative position of the token is more critical to the understanding of the sequence of text than the exact positions [43]. Therefore, for larger input sequences, the grouping does not interfere with the method's performance. The formula to compute relative positional encoding for the positional encoding matrix $P \in \mathbb{R}^{B \times L_{\text{seq}}}$ using grouped attention is given by

$$P_g = \lfloor P / G_s \rfloor$$

where G_s is the group size, B is the batch size and L_{seq} is the sequence length. This maps unseen large positions to values within the pretraining range. The grouped attention computes relative positions and is given by

$$\langle q_m, k_n \rangle_g = g(x_m, x_n, \lfloor (m - n)/G_s \rfloor).$$

The floor operation is used to mitigate the out-of-distribution issue for long input sequences. However, The former method alone hinders the model's performance and reduces the coherence of the generated text. This is because for immediate neighboring tokens, the exact position matters. For instance, "Only one UE connects to the base station." is very different from "One UE only connects to the base station," making it essential to maintain the exact values when looking at immediate neighboring tokens. Therefore, SelfExtend also applies the standard attention to the tokens closest in proximity to the target token. They refer to this as the neighbor attention, given by

$$\langle q_m, k_n \rangle_n = g(x_m, x_n, m-n), \quad \text{if} |m-n| \leq w_n$$

where q_m and k_n represent the transformed embeddings of the query and key tokens respectively. They are derived from the derived from the original token embeddings x_m and x_n. The relative position is expressed as the difference between the absolute positions, m and m, of the query and key tokens. The SelfExtend technique can be represented as

$$\text{Attention}_{m,n} = \begin{cases} \langle q_m, k_n \rangle_n, \text{ if } |m-n| \leq w_n \\ \langle q_m, k_n \rangle_g, \text{ otherwise} \end{cases}$$

where grouped attention $\langle q_m, k_n \rangle_g$ is calculated for key tokens beyond the neighbor window w_n and the neighbor attention $\langle q_m, k_n \rangle_n$ that preserves fine-grained, local dependencies between immediate tokens.

Fig. 7 illustrates the steps taken by SelfExtend to alter the positional encoding. Using SelfExtend, we extend the context window of the pretrained Phi-2's from 2048 tokens to a maximum of 8192. This extension enables better prompting and facilitates the incorporation of tables, a common structured data type in technical documents, and other large data into the model's input, further enhancing its performance.

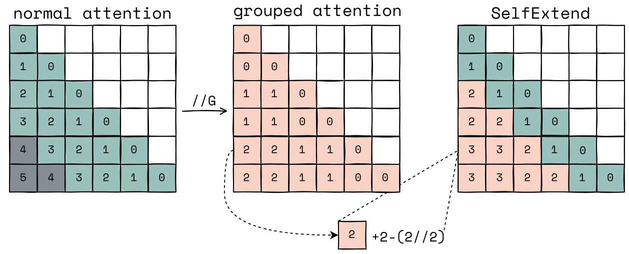


Fig. 7. Illustration of how SelfExtend handles input sequences longer than the pretrained model's context window size. In the figure, the model's context window size, L_{train}, is 4. The neighbor window, w_n is set to 2, and the group size,G_s, is set to 2. Relative positions beyond w_n are calculated using $w_n - \lfloor * \rfloor (w_n/G_s)$.

F. Generator: Fine-Tuned Phi-2 With Multiple Contexts

In this work particular attention has been placed on the use of small, yet efficient, open source models to address the growing computational and privacy challenges in telecommunications. As network complexity increases and edge computing becomes more prevalent, there is a critical need for lightweight models that can perform sophisticated inference tasks without the substantial computational overhead of large language models. Open-source models like Phi-2 offer a promising avenue for developing adaptable, domain-specific solutions that can be fine-tuned with minimal computational resources while maintaining high performance across tasks.

One of the contributors to the LM's inferior performance in specialized technical domains, such as telecom, is the prevalence of technical terms and concepts primarily appearing in white papers and specification documents. To circumvent this, we fine-tune Phi-2, ensuring that the model is adept at adequately leveraging the given context to answer the query and to familiarize it with common terminology terms. To make the fine-tuning process more accessible and reduce the computational and memory demands we utilize gradient accumulation and LoRA [45].

Gradient accumulation enables the model to overcome the memory constraint of training on larger batch sizes by accumulating gradients from multiple smaller batches before updating the weights. The issue with simply using smaller batch sizes is that it could result in noisy updates and affect convergence. Gradient accumulation simulates larger batch sizes, such as 256 which may not be accommodated by the hardware, by creating eight batches of size of size 32 each and updating the model's weights with the accumulated gradient.

Though pretrained LMs might have a large number of parameters, it has been demonstrated that the essential information is concentrated in a subset of these parameters, which is referred to as the model's "intrinsic dimension" [46]. The low intrinsic dimension of models enables the use of techniques, such as LoRA to focus on and efficiently update this core subset during fine-tuning. During fine-tuning, weight updates W_{new} are represented as

$$W_{\text{new}} = W_0 + \Delta W \quad (2)$$

where $W_0 \in \mathcal{R}^{d \times k}$ are the initial pretrained weights, and $\Delta W \in \mathcal{R}^{d \times k}$ represents the change in weights. Computing ΔW is resource-intensive due to the large scale of the matrices involved and the intensive matrix multiplications required for each weight update. In LoRA, trainable parameters ΔW are expressed as a product of two low-rank matrices, $B \times A$, where $B \in \mathcal{R}^{d \times r}$ and $A \in \mathcal{R}^{r \times k}$, with rank $r \ll \min(d, k)$. Thus, with LoRA, weight updates W_{new} are computed as

$$W_{\text{new}} = W_0 + \left(\frac{\alpha}{r}\right) B \times A. \quad (3)$$

Parameter α serves as a scaling factor, reflecting the impact of the weight updates on the initial pretrained weights. Therefore, due to this matrix decomposition, only $d \times r + r \times k$ parameters require updating.

Though other techniques, such as using adapter layers [47], also seek to reduce the number of parameters trained during fine-tuning and make the process more efficient, They often introduce additional latency at inference time. This is not the case with LoRA. In LoRA, the optimized weights are merged with the frozen weights of the base model during deployment, ensuring efficient parameter utilization without additional latency. Real-time processing is essential for telecom applications given the time-sensitive nature of tasks, such as network monitoring, fraud detection, and customer support, for which this work is a precursor.

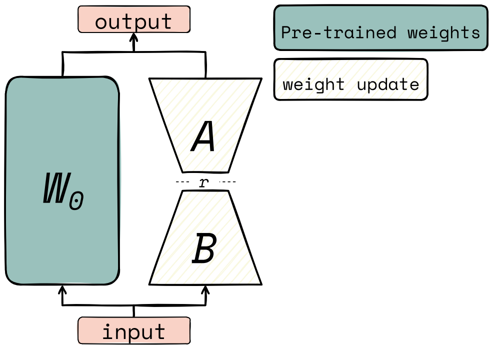


Fig. 8. Schematic illustration of the Low-Rank Adaptation (LoRA) technique for efficient fine-tuning of neural networks with low-rank matrices (e.g., LMs).

Quantized low-rank adaptation (QLoRA) is variation of LoRA in which the model is quantized to 4-bit precision [48]. QLoRA effectively reduces the model's size, increasing computational efficiency and maintaining low resource requirements, without compromising performance. This is particularly significant because maintaining reliable accuracy is essential for telecom applications. LoRA and its variation QLoRA, offer enhanced improved memory efficiency both in the training and in storing the fine-tuned models. LoRA stores the base model separately from the fine-tuned weights, significantly reducing the memory footprint of the fine-tuned model. This decoupling of the base model from task-specific weights enables efficient storage and streamlined management.

G. Prompt Engineering

To fully leverage the capabilities of SLM, effective prompting must be implemented. In this work, the query is placed in the prompt with a set of instructions and the retrieved relevant context, as seen in Fig. 9. The instructions serve to guide the LM through the steps that it should take before answering the question. They bring attention to critical key terms, such as "main" and "primary" and telecom-specific language which can be instrumental in correctly interpreting the nuances of the question and identifying the most relevant context.

The LM is also guided toward adequately relying on the provided context, ensuring that the model's prior knowledge is less likely to interfere with the language model's response. Lastly, the instructions are also a means to ensure that the outputs of Phi-2 are uniform to ensure the answers can be extracted from the output.

IV. Experimental Results

A. Settings

The RAG framework leverages a database containing approximately 550 semantically chunked 3GPP documents, encompassing specifications up to Release 18. For semantic chunking, a breakpoint percentile threshold of 90, and a buffer size of 3, specifying that three sentences would be grouped as a unit when comparing similarity, and that a breakpoint is created when cosine dissimilarity of 90 is reached.

To finetune the generator, we utilize the TeleQnA dataset [18], which comprises 10 000 multiple-choice questions categorized by topic. Each question entry has up to

```
prompt:
<context>
Provide a correct answer to a multiple choice question on wireless
communications and standards. Use only one option from A, B, C, D or E.
<question>
<options>

Instructions:
Step 1. Read the context carefully to identify the most relevant
information.
Step 2. Pay special attention to key terms in the question, such as
"main," "primary," "most important," etc. These terms indicate that the
answer should focus on the most significant aspect mentioned in the
context.
Step 3. Focus on keywords and phrases in the context that directly relate
to the question being asked.
Step 4. Match the context to the options provided and choose the option
that is most explicitly supported by the context and aligns with the key
terms in the question.
Step 5. If the context does not provide clear support for any option,
choose the option that logically fits the context and the question.
Step 6. Pay attention to abbreviations related to wireless communications
and 3GPP standards'

Answer:
```

Fig. 9. Prompt structure that includes retrieved context and instructions.

five options, a single correct answer, and an accompanying justification for the solution. The testing dataset consists of a set of 2000 questions [49], each entry consisting of a question with and up to five answer options. The type of questions include true/false statements, inquiries about specifications and definitions, and questions regarding the purpose and role of entities.

We also evaluate TeleOracle's performance on the ORAN-Bench-13K MCQ dataset [16]. This dataset contains questions of various three difficulties; easy, medium and hard. Easy questions require basic knowledge of O-RAN, medium questions require some reasoning and calculation, while hard questions necessitate a deeper understanding of the concepts and the ability to link pieces of information. We removed some entries with blank questions resulting an O-RAN MCQ dataset consisting of 1139 easy questions, 9454 medium difficulty questions, and 3226 hard questions.

For fine-tuning, the following hyperparameters are set: learning rate of 10^{-4}, batch size of 32, dropout rate of 0.05, weight decay of 0.01. The rank and alpha for LoRA are 32 and 64, respectively. The hybrid search retriever is configured to retrieve the top 150 relevant chunks and the reranker is set to return the top 15 chunks from the 150 retrieved chunks. To account for the chunk size variance introduced when using a semantic chunker, we employ a second query engine that retrieves extra chunks from a database with fixed-size chunks until a set minimum token size is reached. We perform this to ensure that the context window is fully utilized. The aforementioned hyperparameters are not necessarily optimized, but rather set based on qualitative judgment and limited exploration. For reproducibility and reuse, our source code is made publicly available.[1]

B. Results and Analysis

To examine the performance of TeleOracle we perform tests on multiple datasets, in addition to performing both an ablation study and a full-model evaluation. This provides an in-depth understanding of the model's behavior and effectiveness. In

[1] source code available at https://github.com/Nouf-Alabbasi/oKUmura_AI_Telecom_challenge.

TABLE I
ACCURACY COMPARISON OF OUR FINE-TUNED PHI-2 BASED RAG MODEL AGAINST BASELINE MODELS ON TELECOM STANDARDS QUESTIONS, BOTH WITH AND WITHOUT RETRIEVED CONTEXT

	Accuracy	
Model	**w/o Context**	**w/ Context**
Phi-2	49.95%	71.70%
GPT-4o mini	63.20%	84.20%
GPT-4o	61.30%	82.05%
TeleOracle (QLoRA 4-bits)	-	80.80%
TeleOracle (LoRA)	**-**	**81.20%**

TABLE II
FAITHFULNESS SCORES OF TELEORACLE, GPT-4O MINI, AND GPT-4O. TELEORACLE ACHIEVES THE HIGHEST SCORE OF 78.80%, INDICATING STRONGER ALIGNMENT BETWEEN ITS RESPONSES AND THE RETRIEVED CONTEXT COMPARED TO THE OTHER MODELS

Model	**Faithfulnes s**
GPT-4o mini	78.25%
GPT-4o	76.07%
TeleOracle	**78.80%**

Table I, we benchmark our model against base Phi-2 (2.7B), GPT-4o mini, and GPT4o. We find that the base Phi-2 model considerably underperforms. Similarly, we find that leveraging retrieved context noticeably enhances all the benchmarked models' performance, indicating the efficacy of using RAG in this context.

Notably, GPT-4o achieves only 82.05% accuracy, performing slightly worse than its smaller counterpart. This lower performance can be attributed to its highly generalized nature, which leads to misinterpretation of specialized telecommunications terminology and concepts. To illustrate this knowledge interference phenomenon, consider the following example MCQ question where GPT-4's response reveals how general knowledge can interfere with domain-specific understanding:

How can the impact of noise in MPAC on FR1 MIMO OTA performance be corrected?.

1. By decreasing the attenuation value.

2. By increasing the attenuation value.

3. By replacing the amplifiers.

4. By adjusting the amplifier output power.

Both TeleOracle and GPT-4o mini pick the correct answer choice, namely option 2. However, GPT-4o choses option 4 which states that the noise impact in MPAC (MultiProbe Anechoic Chamber) on FR1 multiple-input multiple-output (MIMO) OTA (Over-The-Air) performance can be corrected "*by adjusting the amplifier output power*". This indicates that GPT-4o is relying on its acquired general knowledge and is conflating engineering principles with telecom-specific principles. In general engineering principles, increasing signal power is a common method to improve performance in noisy environments. However, in the specific context of MPAC testing for FR1 MIMO OTA performance, increasing the attenuation value is the correct method to mitigate the impact of noise.

We quantify this discrepancy using the faithfulness metric, which measures the extent to which the model's response aligns with the retrieved context [28]. A higher faithfulness indicates stronger alignment between the context and the generated output. In telecom, where technical precision is paramount, this alignment directly influences the reliability and accuracy of responses. As shown in Table II, the evaluation results highlight that GPT-4o underperforms compared to both TeleOracle and GPT-4o mini, underscoring its reduced ability to effectively utilize the retrieved context, a key factor for accurate telecom-specific answers.

Furthermore, the results in Wu et al. [50] work suggest that there is a relation between the model's confidence in its prior knowledge and adherence to the retrieved context. This further reinforces our hypothesis that aligning the model through fine-tuning is crucial for improving its accuracy and reducing potential conflations. Additionally, despite clear instructions provided, GPT4-o does not consistently adhere to the required output format, leading to the predicted answer option not being extracted properly. It is likely that larger models, such as GPT-4o, overfit on generating free-form, coherent text, making them less responsive to structured output requirements. This further underscores the importance of fine-tuning the model to overcome these challenges, given that adhering to the required format holds extreme significance in certain downstream applications in telecom.

We also note that the model fine-tuned using QLoRA achieves comparable performance to that of the model fine-tuned using LoRA with accuracies 80.80% and 81.20% respectively. Quantizing a model during fine-tuning can save memory and computational resources, but it can also introduce small inaccuracies in the model's representations, leading to a minor drop in performance. The fact that this degradation is only slight indicates that QLoRA and LoRA are robust to quantization, maintaining much of their effectiveness even in a more compressed form. From these results, it is evident that fine-tuning the SLM and utilizing our RAG framework are effective techniques to align Phi-2 to the telecom domain.

To further analyze the results and the impact of each involved component in the architecture, we perform an ablation study. This analysis reveals that Re-ranking, with an accuracy of 77%, contributes significantly to the overall performance improvement. The findings underscore the importance of a two-stage retrieval process, which effectively captures the intricate relation between the query and context chunks while maintaining an acceptable retrieval speed. While the addition of the hybrid search retriever has a marginal effect on the base model's accuracy (only improving accuracy by around 1%), it plays a crucial role by capturing important telecom-specific keywords within the retrieved context chunks that the vector retriever alone might overlook, as demonstrated in Fig. 3.

It is significant to point out that the impact of selfExtend becomes more pronounced when semantic chunking is used resulting in an 8% increase compared to the base model, raising the accuracy to 80.30%. Semantic chunking produces

TABLE III
PERFORMANCE COMPARISON OF VARIOUS CONFIGURATIONS OF THE FINE-TUNED PHI-2 MODEL WITH RAG AND ADDITIONAL COMPONENTS ON TELECOM SPECIFICATION QNA. THE TABLE USES THE FOLLOWING ACRONYMS: SE FOR SELFEXTEND, RR FOR RERANK, SC FOR SEMANTIC CHUNKING, AND MC FOR MULTIPLE CONTEXT

Model	**Accuracy**
FT Phi-2 + VR	72.10%
FT Phi-2 + VR + RR	77.35%
FT Phi-2 + VR + RR + SC	41.20%
FT Phi-2 + VR + RR + SC + SE	80.30%
FT Phi-2 + HR + RR + SC + SE	81.10%
FT Phi-2 (MC) + HR + RR + SC + SE	81.20%

TABLE IV
ACCURACY COMPARISON OF DIFFERENT MODELS WITH CONTEXT ON MULTIPLE MCQ DATASETS. THE O-RAN MCQ DATASET VARIANTS INCLUDE ORAN-E (EASY), ORAN-M (MEDIUM), AND ORAN-H (HARD), CATEGORIZED BASED ON THE DIFFICULTY LEVELS OF THE QUESTIONS

	TeleQnA	**ORAN-E**	**ORAN-M**	**ORAN-H**
TeleOracle	81.20%	79.80%	76.37%	73.34%
GPT-4o mini	84.20%	80.77%	76.21%	72.78%
Phi-2	71.35%	75.33%	72.79%	69.53%

more cohesive chunks, although they tend to be longer, as shown in Fig. 2. SelfExtend extends the context window to allow the incorporation of these chunks into the prompt while ensuring that critical information is retained. We also observe the effect of fine-tuning with multiple retrieved contexts in the prompt. This enhancement, illustrated by the slight improvement in accuracy, can be attributed to the model's improved ability to leverage nuanced ideas from the context and draw more accurate inferences.

While TeleOracle was finetuned on 3GPP MCQs, it demonstrated strong performance on the ORAN-Bench-13K MCQ dataset [16] without additional finetuning, highlighting the model's cross-domain generalization capabilities, as shown in Table IV. On ORAN-E (easy difficulty), TeleOracle achieved 79.80% accuracy without additional fine-tuning. This performance is notably better than that of the base Phi-2 model, showing the model's robust transfer capabilities to telecom-specific question answering. The O-RAN dataset results show a gradual performance decrease as question difficulty increases, with accuracies of 76.37% and 73.34% for ORAN-M and ORAN-H, respectively. This expected decline reflects the increasing complexity of the questions while still maintaining strong performance across all difficulty levels. Notably, even for the most challenging questions in ORAN-H, TeleOracle maintains an accuracy above 73%, demonstrating robust generalization despite having no exposure to O-RAN specific training data. TeleOracle's performance is comparable to that of proprietary models, such as GPT-4o mini, which achieves 72.78% in ORAN-H questions. These comprehensive results demonstrate TeleOracle's exceptional adaptability across diverse telecommunication domains, offering a robust question-answering solution that eliminates the need for additional fine-tuning on specialized datasets.

We leveraged a semantic router[10] to enable the model to alternate between DBs automatically given a query and the results from our test illustrate its effectiveness in routing the query to the appropriate database. The semantic router achieved an accuracy of 91.18% without optimization. After optimization of the threshold, the router achieves an accuracy of 93.99%

V. CONCLUSIONS AND FUTURE WORKS

The complex nature of telecom documentation necessitates specialized language models that can handle the domain's nuances. In this work, we present TeleOracle, an advanced RAG framework that leverages SLMs for telecommunications tasks. By integrating LoRA, context window extension, semantic chunking, and a two-stage retrieval process, our system efficiently processes complex documentation while maintaining computational efficiency. Our experimental results demonstrate that TeleOracle outperforms baseline models in both accuracy and faithfulness, demonstrating the viability of specialized SLMs for telecommunications applications and establishing a foundation for efficient language-based solutions in the field.

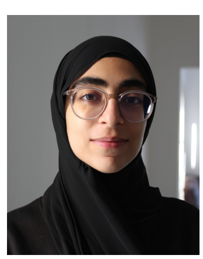
A significant limitation of RAG models is their inability to effectively incorporate tables, figures, network graphs, and communication-specific features, such as channel information [51]. These data types contain valuable information that can enhance the LM's responses. Future work can focus on integrating these elements into the retrieved context. Beyond data integration challenges, certain questions and tasks require careful reasoning and planning. For example, queries like 'How can the UPF assist RAN with identifying the end of DL traffic bursts?' demand complex analysis that may exceed the capabilities of one-shot prompting. This limitation highlights the need to explore advanced reasoning techniques and domain-specific approaches.

Furthermore, TeleOracle does not have access to user data, channel state information (CSI), or network topology, which limits its ability to adapt dynamically to real-time network conditions. More advanced use cases can potentially benefit from incorporating these elements, but how to effectively integrate them within this framework remains an open research question. Exploring this direction paves the way for agentic RAG and broader applications in optimization problems across multiple layers of the network stack. Potential areas of investigation include adaptive beamforming and precoding in MIMO systems, intelligent resource allocation, interference management in dense networks, network slicing optimization, and energy-efficient wireless networking.

REFERENCES

[1] O. Erak et al., "Leveraging fine-tuned retrieval-augmented generation with long-context support: For 3GPP standards," in *Proc. IEEE Globecom Workshops*, 2025, pp. 1–6.
[2] O. Erak, O. Alhussein, S. Naser, N. Alabbasi, D. Mi, and S. Muhaidat, "Large language model-driven curriculum design for mobile networks," 2024, *arXiv:2405.18039*.

[3] A. Maatouk, N. Piovesan, F. Ayed, A. D. Domenico, and M. Debbah, "Large language models for telecom: Forthcoming impact on the industry," *IEEE Commun. Mag.*, vol. 63, no. 1, pp. 62–68, Jan. 2025.
[4] R. Zhang et al., "Interactive AI with retrieval-augmented generation for next generation networking," *IEEE Netw.*, vol. 38, no. 6, pp. 414–424, Nov. 2024.
[5] X. Lin et al., "A primer on generative AI for telecom: From theory to practice," 2024, *arXiv:2408.09031*.
[6] S. Soman and G. H. Ranjani, "Observations on LLMs for telecom domain: Capabilities and limitations," in *Proc. Int. Conf. AI-ML Syst.*, 2023, pp. 1–5.
[7] H. Zhou et al., "Large language model (LLM) for telecommunications: A comprehensive survey on principles, key techniques, and opportunities," 2024, *arXiv:2405.10825*.
[8] N. Zhong et al., "CASIT: Collective intelligent agent system for Internet of Things," *IEEE Internet Things J.*, vol. 11, no. 11, pp. 19646–19656, Jun. 2024.
[9] R. Xie et al., "Large language model-aided edge learning in distribution system state estimation," *IEEE Internet Things J.*, early access, Oct. 18, 2024, doi: 10.1109/JIOT.2024.3483441.
[10] "Semantic-router: Naurelio-labs." 2024. [Online]. Available: https://github.com/aurelio-labs/semantic-router
[11] M. Javaheripi and S. Bubeck (Microsoft, Redmond, WA, USA). *Phi-2: The Surprising Power of Small Language Models*. Dec. 2023. Accessed: Aug. 19, 2024. [Online]. Available: https://www.microsoft.com/en-us/research/blog/phi-2-the-surprising-power-of-small-language-models/
[12] R. Anil et al., "Gemini: A family of highly capable multimodal models," 2024, *arXiv:2312.11805*.
[13] Y. Yao, X. Xu, and Y. Liu, "Large language model unlearning," 2024, *arXiv:2310.10683*.
[14] I. Karim, K. S. Mubasshir, M. M. Rahman, and E. Bertino, "SPEC5G: A dataset for 5G cellular network protocol analysis," in *Proc. IJCNLP-AACL*, 2023, pp. 20–38.
[15] R. Nikbakht, M. Benzaghta, and G. Geraci, "TSpec-LLM: An open-source dataset for LLM understanding of 3GPP specifications," 2024, *arXiv:2406.01768*.
[16] P. Gajjar and V. K. Shah, "ORAN-Bench-13K: An open source benchmark for assessing LLMs in open radio access networks," 2024, *arXiv:2407.06245*.
[17] H. Holm, "Bidirectional encoder representations from transformers (BERT) for question answering in the telecom domain.: Adapting a BERT-like language model to the telecom domain using the ELECTRA pre-training approach," M.S. thesis, KTH Royal Inst. Technol., Stockholm, Sweden, 2021. [Online]. Available: http://urn.kb.se/resolve?urn=urn:nbn:se:kth:diva-301313
[18] A. Maatouk, F. Ayed, N. Piovesan, A. D. Domenico, M. Debbah, and Z.-Q. Luo, "TeleQnA: A benchmark dataset to assess large language models telecommunications knowledge," 2023, *arXiv:2310.15051*.
[19] T. Ahmed, N. Piovesan, A. D. Domenico, and S. Choudhury, "Linguistic intelligence in large language models for telecommunications," 2024, *arXiv:2402.15818*.
[20] L. Bariah, H. Zou, Q. Zhao, B. Mouhouche, F. Bader, and M. Debbah, "Understanding telecom language through large language models," in *Proc. IEEE Globecom*, 2023, pp. 6542–6547.
[21] A. Karapantelakis et al., "Using large language models to understand telecom standards," 2024, *arXiv:2404.02929*.
[22] A.-L. Bornea, F. Ayed, A. D. Domenico, N. Piovesan, and A. Maatouk, "Telco-RAG: Navigating the challenges of retrieval-augmented language models for telecommunications," 2024, *arXiv:2404.15939*.
[23] G. M. Yilma, J. A. Ayala-Romero, A. Garcia-Saavedra, and X. Costa-Perez, "TelecomRAG: Taming telecom standards with retrieval augmented generation and LLMs," 2024, *arXiv:2406.07053*.
[24] N. Piovesan, A. D. Domenico, and F. Ayed, "Telecom language models: Must they be large?" 2024, *arXiv:2403.04666*.
[25] B. Guda, G. Z. Ashungafac, L. Francis, and C. Joe-Wong, "QMOS: Enhancing LLMs for telecommunication with question masked loss and option shuffling," 2024, *arXiv:2409.14175*.
[26] G. C. Lee, D. Khu, F. Guretno, and E. Kurniawan, "Specializing language models for 3GPP standards: Enhancements for technical document queries," in *Proc. IEEE Globecom Workshops*, 2024.
[27] A. Maatouk, K. C. Ampudia, R. Ying, and L. Tassiulas, "Tele-LLMs: A series of Specialized large language models for telecommunications," 2024, *arXiv:2409.05314*.
[28] S. Es, J. James, L. Espinosa-Anke, and S. Schockaert, "RAGAS: Automated evaluation of retrieval augmented generation," 2023, *arXiv:2309.15217*.
[29] S. Roychowdhury, S. Soman, H. G. Ranjani, N. Gunda, V. Chhabra, and S. K. Bala, "Evaluation of RAG metrics for question answering in the telecom domain," 2024, *arXiv:2407.12873*.
[30] Q. Wu, B. Shen, and Y. Chen, "Learning to recommend trigger-action rules for end-user development: A knowledge graph based approach," in *Proc. 19th Int. Conf. Softw. Syst. Reuse*, Dec. 2020, pp. 190–207.
[31] Z. Huang, J. Li, H. Zhang, C. Zhang, and G. Yu, "TAP-AHGNN: An attention-based heterogeneous graph neural network for service recommendation on trigger-action programming platform," in *Proc. Int. Conf. Intell. Comput.*, Jul. 2023, pp. 141–152.
[32] Z. Kuang, X. Xiong, G. Wu, F. Wang, J. Zhao, and D. Sun, "A recommendation system for trigger–action programming rules via graph contrastive learning," *Sensors*, vol. 24, no. 18, p. 6151, Sep. 2024.
[33] Y. Cheng, M. Xu, Y. Zhang, K. Li, R. Wang, and L. Yang, "AutoIoT: Automated IoT platform using large language models," *IEEE Internet Things J.*, early access, Dec. 30, 2024, doi: 10.1109/JIOT.2024.3523907.
[34] S. Xiao, Z. Liu, P. Zhang, N. Muennighoff, D. Lian, and J.-Y. Nie, "C-Pack: Packaged resources to advance general Chinese embedding," 2024, *arXiv:2309.07597*.
[35] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. ICML*, 2020, pp. 1597–1607.
[36] A. Askari, A. Abolghasemi, G. Pasi, W. Kraaij, and S. Verberne, "Injecting the BM25 score as text improves BERT-based re-rankers," in *Proc. Eur. Conf. Inf. Retr.*, 2023, pp. 66–83.
[37] S. Bruch, S. Gai, and A. Ingber, "An analysis of fusion functions for hybrid retrieval," *ACM Trans. Inf. Syst.*, vol. 42, no. 1, pp. 1–35, 2022.
[38] N. F. Liu et al., "Lost in the middle: How language models use long contexts," 2023, *arXiv:2307.03172*.
[39] (Inferless, Bengaluru, India). *MS MARCO: Ms-Marco-MINILM-l-6-v2*. 2023. Accessed: Aug. 18, 2024. [Online]. Available: https://huggingface.co/cross-encoder/ms-marco-MiniLM-L-6-v2
[40] S. Chen, S. Wong, L. Chen, and Y. Tian, "Extending context window of large language models via positional interpolation," 2023, *arXiv:2306.15595*.
[41] G. Chen, X. Li, Z. Meng, S. Liang, and L. Bing, "CLEX: Continuous length extrapolation for large language models," 2024, *arXiv:2310.16450*.
[42] B. Peng, J. Quesnelle, H. Fan, and E. Shippole, "YaRN: Efficient context window extension of large language models," 2023, *arXiv:2309.00071*.
[43] H. Jin et al., "LLM maybe LongLM: Self-extend LLM context window without tuning," 2024, *arXiv:2401.01325*.
[44] J. Su, Y. Lu, S. Pan, A. Murtadha, B. Wen, and Y. Liu, "RoFormer: Enhanced transformer with rotary position embedding," 2023, *arXiv:2104.09864*.
[45] E. J. Hu et al., "LoRA: Low-rank adaptation of large language models," in *Proc. ICLR*, 2022, pp. 1–26.
[46] A. Aghajanyan, S. Gupta, and L. Zettlemoyer, "Intrinsic dimensionality explains the effectiveness of language model fine-tuning," in *Proc. ACL-IJCNLP*, 2021, pp. 7319–7328.
[47] N. Houlsby et al., "Parameter-efficient transfer learning for NLP," 2019, *arXiv:1902.00751*.
[48] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient finetuning of quantized LLMs," 2023, *arXiv:2305.14314*.
[49] "Specializing large language models for telecom networks by ITU AI/ML in 5G challenge." Accessed: Jul. 10, 2024. [Online]. Available: https://zindi.africa/competitions/specializing-large-language-models-for-telecom-networks
[50] K. Wu, E. Wu, and J. Zou, "ClashEval: Quantifying the tug-of-war between an LLM's internal prior and external evidence," 2024, *arXiv:2404.10198*.
[51] X. Meng, C. Lin, Y. Wang, and Y. Zhang, "NetGPT: Generative pretrained transformer for network traffic," 2023, *arXiv:2304.09513*.



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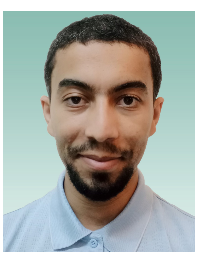

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